A PROJECT REPORT ON

AIR DEFENCE AND TARGETING SYSTEM

Submitted in partial fulfillment of the requirements for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

ELECTRONICS AND COMMUNICATION ENGINEERING

Under the guidance of

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(Autonomous – NBA Accredited)

(Approved by AICTE, New Delhi & Affiliated to JNTUA, Ananthapuramu)

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

(2025-2026)

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This is to certify that the project work entitled “**AIR DEFENCE AND TARGETING SYSTEM”** is a genuine work of

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

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IM1: Unleash and encourage the innate potential and creativity of students.

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PEO1:Have Professional competency through the application of knowledge gained from subjects like Mathematics, Physics, Chemistry, Inter-Disciplinary and core subjects like Signal Processing, VLSI, Embedded Systems, Communication and Automation.**(Professional Competency)**

PEO2: *Excel in one's career by critical thinking towards successful services and growth of the organization or as an entrepreneur or through higher studies.* **(Successful Career Goals)**

PEO3: Enhance knowledge by updating *advanced technological concepts* for facing the *rapidly changing world and contribute to society through innovation and creativity.* **(Continuing Education and Contribution to Society)**

PROGRAMME OUTCOMES (PO's)

On Successful completion, the graduate will be able to,

PO1. **Engineering knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. **Problem analysis**: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. **Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

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PO8. **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. **Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. **Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. **Project management and finance**: Demonstrate knowledge and understanding of the Engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context Technological change.

PROGRAM SPECIFIC OUTCOMES (PSO's)

On Successful completion, the graduate will be able to,

PSO1: Apply the knowledge obtained in core areas for the analysis and design of components in Signal Processing, VLSI, Embedded Systems, Communication and Automation.

PSO2: Adapt Innovation, Creativity and design to develop products which meet industrial and societal needs.

Course Outcomes for project work

On completion of project work we will be able to,

CO1. Demonstrate in-depth knowledge of embedded systems by developing an air defence and targeting system using ESP32, ultrasonic sensors, and servo motors.

CO2. Identify and analyze the problem of real-time object detection and tracking and formulate a solution using ultrasonic sensing and microcontroller-based control logic.

CO3. Design and implement a working system capable of detecting objects and accurately positioning servos for target alignment and laser indication.

CO4. Undertake investigation of ultrasonic sensors to measure distance and validate system performance under different conditions to ensure reliable operation to provide valid conclusions.

CO5. Use appropriate engineering tools such as Arduino IDE, embedded C programming, and electronic components (ESP32, sensors, servos) for system development.

CO6. Apply the developed system concept to real-world applications such as surveillance, security, and automated detection systems contributing to societal safety.

CO7. Analyze the environmental impact of the system, focusing on energy efficiency, minimal hardware usage, and safe operation of electronic components.

CO8. Understand professional and ethical considerations, ensuring safe use of laser modules and responsible application of detection and targeting systems.

CO9. Function effectively as individual and a member in the project team.

CO10. Develop communication skills, both oral and written for preparing and presenting project report.

CO11. Demonstrate knowledge and understanding of cost and time analysis required for carrying out the project.

CO12. Engage in lifelong learning to improve knowledge and competence in the chosen area of the project.

CO – PO MAPPING

CO\PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO.1	3	-	-	-	-	-	-	-	-	-	-	-	3	3
CO.2	-	3	-	-	-	-	-	-	-	-	-	-	3	3
CO.3	-	-	3	-	-	-	-	-	-	-	-	-	3	3
CO.4	-	-	-	3	-	-	-	-	-	-	-	-	3	3
CO.5	-	-	-	-	3	-	-	-	-	-	-	-	3	3
CO.6	-	-	-	-	-	3	-	-	-	-	-	-	3	3
CO.7	-	-	-	-	-	-	3	-	-	-	-	-	3	3
CO.8	-	-	-	-	-	-	-	3	-	-	-	-	3	3
CO.9	-	-	-	-	-	-	-	-	3	-	-	-	3	3
CO.10	-	-	-	-	-	-	-	-	-	3	-	-	3	3
CO.11	-	-	-	-	-	-	-	-	-	-	3	-	3	3
CO.12	-	-	-	-	-	-	-	-	-	-	-	3	3	3
CO	3	3	3	3	3	3	3	3	3	3	3	3	3	3

ABSTRACT

The primary objective of this project is to reduce the difficulties encountered This project offers an Air Defense and Targeting System that uses an inexpensive embedded setup to identify and react to nearby flying objects. The device measures an object's distance in real time using an ultrasonic sensor that functions as a short-range radar. To scan a larger region and determine the target's direction, a servo motor system turns the sensor. The ESP32 controller examines the angle and distance data and initiates the targeting operation when an item moves within the designated danger zone. The servo motor simulates an automatic aiming procedure by aligning a laser module toward the target based on the detected position. The suggested system shows how the fundamental operation of an air defense monitoring and response model can be replicated using sensing, scanning, and automated aiming. Compact and inexpensive, this prototype can be expanded with better tracking algorithms and wifi monitoring.

Keywords : ESP32, Ultrasonic Sensor, Servo Motor, Laser Targeting, Air defence System.

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INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

A significant change in the dynamics of international security is currently being led by the field of Electronics and Communication Engineering (ECE). In the past, air defence, which relied on enormous radar arrays and long-range interceptors, was the sole purview of gigantic military industrial complexes. But one major technological difficulty of the modern period is the emergence of "asymmetric aerial threats." These unmanned aerial vehicles (UAVs) can avoid conventional high-altitude defence grids because they are small, nimble, and incredibly inexpensive. The necessity to utilise fundamental engineering concepts—signal processing, embedded systems, and automated control—to develop a localised, economical solution to this contemporary issue is what inspired us as ECE students to work on this project, Air Defence and Targeting System.

This study's context is the susceptibility of some high-value areas, like field outposts or localised infrastructure, to short-range invasions. For these situations, conventional defence systems are frequently "over-engineered"; tracking a hobbyist-grade drone with a high-precision missile is not financially feasible. By proving that available embedded technology can effectively miniaturise a "detect-to-engage" loop, this initiative aims to close that gap. We can interpret real-time geographical data to maintain a protective "danger zone" surrounding a sensitive region by using an ESP32 controller as the central intelligence unit.

Our main goal is to investigate how sensor technology and mechanical actuation might work together. We reproduce the basic mechanics of target acquisition for a fraction of the conventional cost by combining a servo motor system for wide-area scanning with an ultrasonic sensor to serve as a short-range radar. The fundamental objective of any defence model is represented by the system's capacity to go from passive scanning to active laser targeting: converting unprocessed data into an automated, useful reaction.

The goal of this project is to demonstrate that advanced defence logic—scanning, identifying, and aiming—is no longer limited to pricey hardware. It provides a scalable basis for the upcoming generation of reasonably priced, automated security systems by acting as a prototype for upcoming ECE-driven improvements like wireless monitoring and sophisticated tracking algorithms.

1.2 SYSTEM ASSEMBLY AND CORE EMBEDDED ARCHITECTURE

The Air Defence and Targeting System's core assembly is an intricate combination of precise electromechanical control and fast digital processing. The ESP32 microcontroller, a potent dual-core processor chosen for its high clock speed and integrated Wi-Fi/Bluetooth capabilities—both crucial for real-time defence logic—is at the centre of this architecture. In contrast to conventional 8-bit controllers, the ESP32 can handle complicated PWM (Pulse Width Modulation) signals for the mechanical actuators while concurrently processing multidirectional sensor data.The "detect-to-engage" loop is formed by the physical and logical division of the system into three main functional layers:

The Sensing Layer makes use of three HC-SR04 ultrasonic sensors arranged in a tri-directional array. These sensors function as a short-range acoustic radar system that monitors the time-of-flight of sound waves to determine target distance. They are positioned strategically to cover the Left, Centre, and Right sectors.

The Actuation Layer: The system uses a 4-servo motor mechanism to offer complete spatial coverage. Smooth and fluid tracking of moving objects is ensured by this multi-servo design, which permits independent control over the sensors' scanning and the targeting module's precise alignment.

The Targeting Layer: A high-intensity laser module is the core assembly's ultimate output. When the system successfully locks onto a target inside the designated danger zone, the laser, which is mounted atop the secondary gimbal system, acts as a visual surrogate for a defence countermeasure and provides instant feedback.

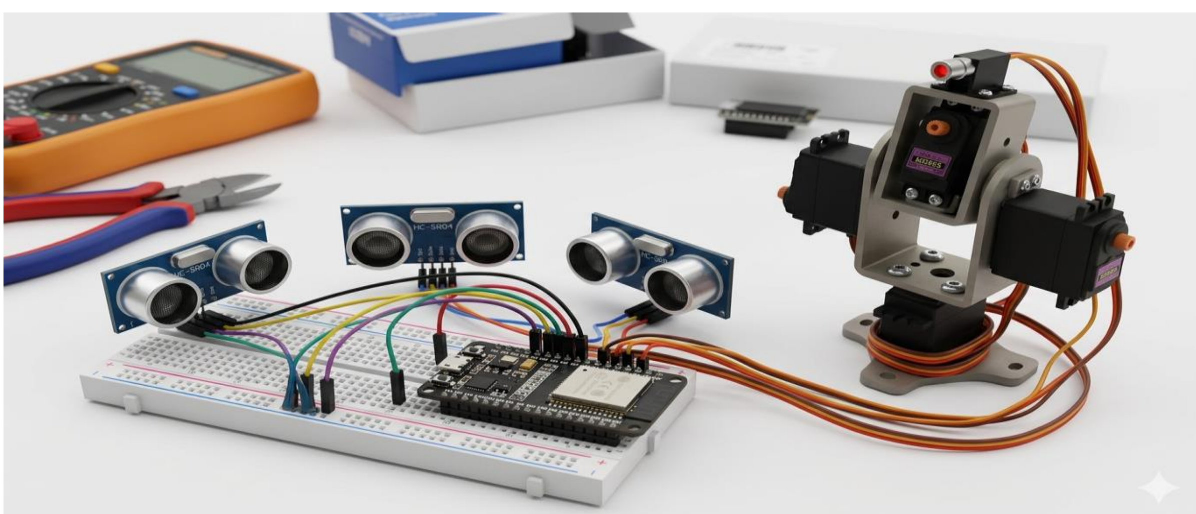

Fig 1.2 System Assembly

1.3 PROJECT RESEARCH AND DEVELOPMENT SCOPE

The goal of this project is to create a scalable, affordable prototype that illustrates the basic technical concepts utilised in contemporary automated defence systems. The demand for low-cost, localised surveillance systems has increased as airborne dangers become smaller and more accessible. The theoretical ideas of Electronics and Communication Engineering (ECE) are connected to real-world hardware implementation through this project.The current work's boundaries and areas of concentration are as follows:

Multi-Directional Spatial Mapping: Using a sensor fusion technique to identify target coordinates, the monitored region is divided into sectors (Left, Center, and Right) using a logic-based approach.

Real-Time Embedded Processing: Reducing delay between the detection of an object and the mechanical tracking's response by optimising the ESP32 firmware.Developing the software logic needed to assess distances from various sensors and automatically identify the most serious threat for interaction is known as automated tracking algorithms.

1.4 DEFENCE SYSTEM LIMITATIONS IDENTIFICATION

It was vital to determine the serious flaws in the current manual and entry-level automated security solutions before creating the suggested system. This project attempts to address the excessive latency and poor situational awareness of many accessible defence models.The following are the main issues found during the research phase:

Inaccuracy of Manual Tracking: Without considerable delay and error, humans are unable to reliably and correctly monitor fast-moving objects in real-time.

Economic Prohibitive Ness of Military Radar: To protect small-scale civilian or localised military assets, professional-grade radar systems are too costly and complicated.

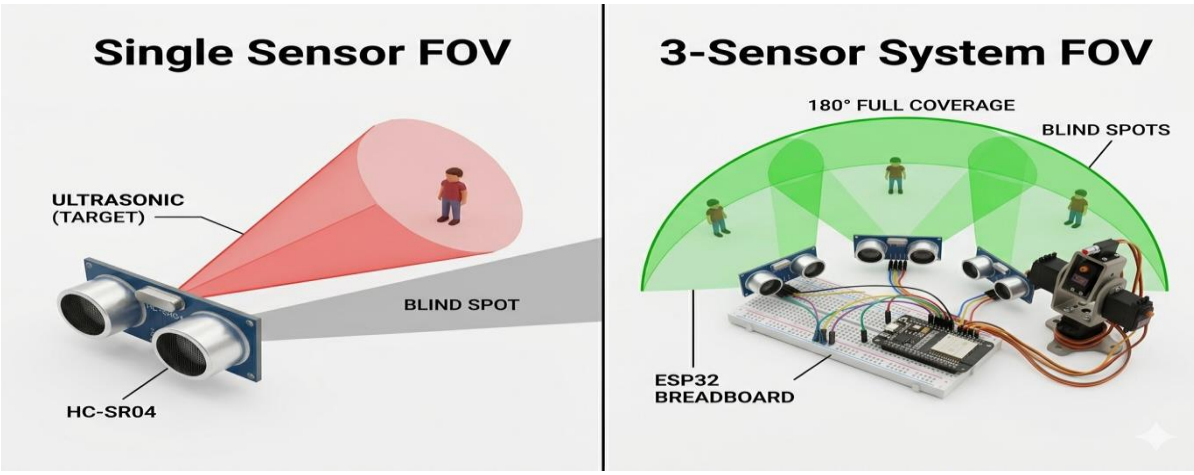

Fig 1.4 Defence System Identification

Single-Sensor Blind Spots: Conventional "radar" systems frequently employ a single sensor that revolves on a motor, resulting in "blind windows" where a target can evade the system when the sensor is pointing in the opposite direction.

Detection without Reaction: A lot of security systems in use today are passive; they might identify an incursion, but they don't have the built-in mechanical capacity to instantly "point" or "target" the threat.

1.5 DESIGN GOALS AND RESEARCH OBJECTIVES

The Air Defence and Targeting System's goal is to offer a thorough, automated solution to the issues found in earlier defence models. The project's goal is to switch from basic detection to an active "Search and Track" mode of operation.The engineering objectives are: Enhanced Situational Awareness:

To minimise the blind areas typical of single-sensor systems, three ultrasonic sensors will be used to offer 180^-degree continuous monitoring. Precision Targeting: Using a high-resolution 4-servo motor array to aim a laser module straight at the identified target to achieve precise mechanical alignment.

High-Speed Control Loop: To make use of the ESP32's processing capacity to guarantee that every aspect of the system, from aiming to detection, runs as smoothly as possible.

Visual Validation: Using a laser indication to clearly demonstrate the tracking accuracy of the device.

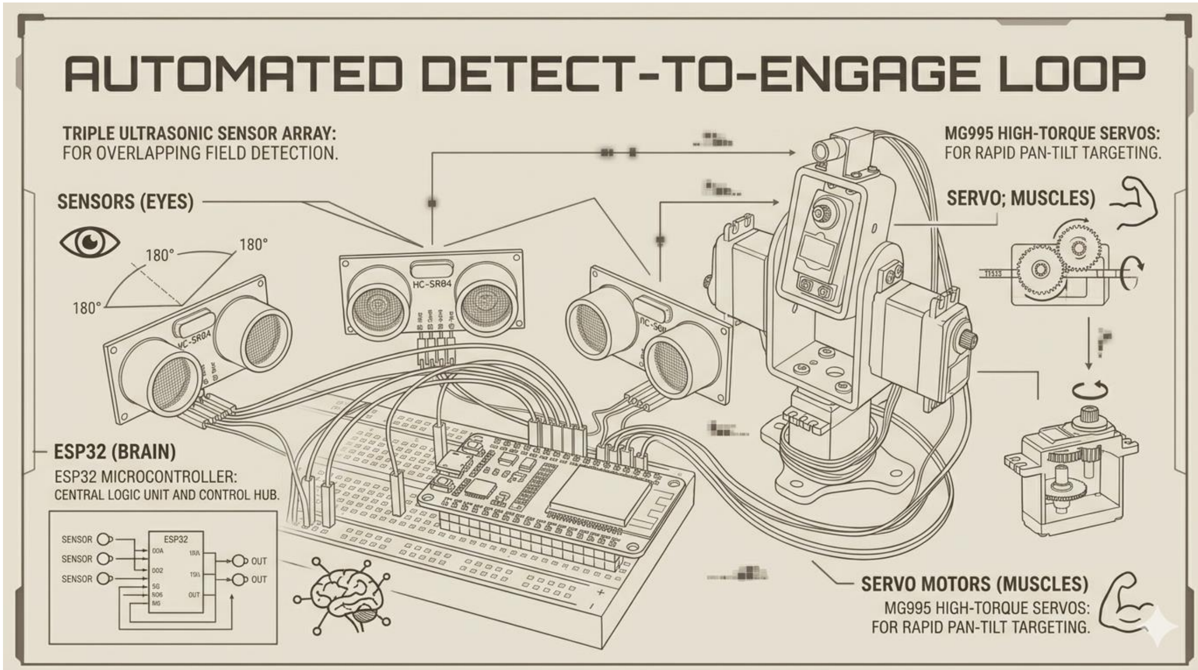

Fig 1.5 Design Goals

1.6 METHODS AND EXECUTION OF THE PROJECT

To guarantee dependability and speed, the methodology used for this project adheres to a strict "Sense-Process-Act" paradigm. The ESP32 controls every phase of the cycle to keep the hardware components in sync.The following are the steps involved in execution:

Continuous Acoustic Scanning: To map the surroundings, the system starts by simultaneously firing ultrasonic pulses from the Left, Center, and Right sensors.

Threshold Validation: After receiving the echo timing, the ESP32 determines how far away any objects are. If an object is within the "Danger Zone" threshold, the tracking sequence starts.

Target Prioritisation: Based on the smallest recorded distance, the system determines the direction of the nearest item by comparing the data from all three sensors.

Synchronised Actuation: The four servo motors rotate and tilt the entire targeting assembly into place once the ESP32 determines the required angles and transmits high-speed PWM signals to them.

Active Engagement Simulation: As long as the item is detected, the system keeps modifying the servos in real-time to maintain the aim. The laser module is activated to signify a "lock" on the target.

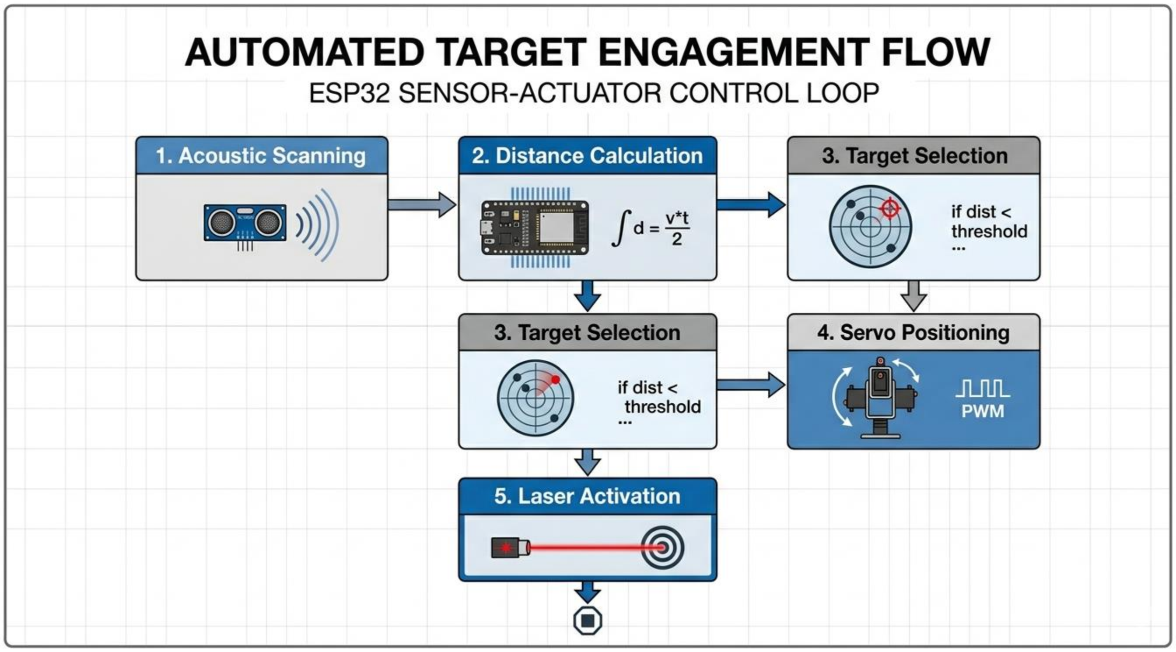

Fig 1.6 Automated Target Engagement Flow

CHAPTER – 2

LITERATURE SURVEY

2.1 EVOLUTION AND MODERN CHALLENGES OF SHORT-RANGE AIR DEFENSE (SHORAD)

Using large radar sites and costly interceptor batteries, the historical structure of air defence was largely intended to fight high-altitude, long-range threats like manned bombers and ballistic missiles. However, the rise of "Low-Slow-Small" (LSS) aerial threats, particularly unmanned aerial vehicles (UAVs), has drastically changed the current security environment. Because of their narrow radar cross-section (RCS), which enables them to fly below the minimum detection height of long-range sensors or blend in with ground clutter, these assets are challenging for conventional defence grids to identify.

This change has required the development of localised, flexible, and affordable Short-Range Air Defence (SHORAD) systems in the discipline of Electronics and Communication Engineering (ECE). The "cost-imposition asymmetry"the unsustainable practice of utilising a million-dollar missile to neutralise a thousand-dollar drone is the fundamental problem with contemporary SHORAD. As a result, current research focuses on creating automated, embedded solutions that can use high-speed microcontrollers to create a "danger zone" perimeter around vital infrastructure.These systems have undergone three major stages of evolution:

Using human operators to visually identify and monitor targets is known as manual surveillance, and it is intrinsically constrained by human reaction times and high error rates.

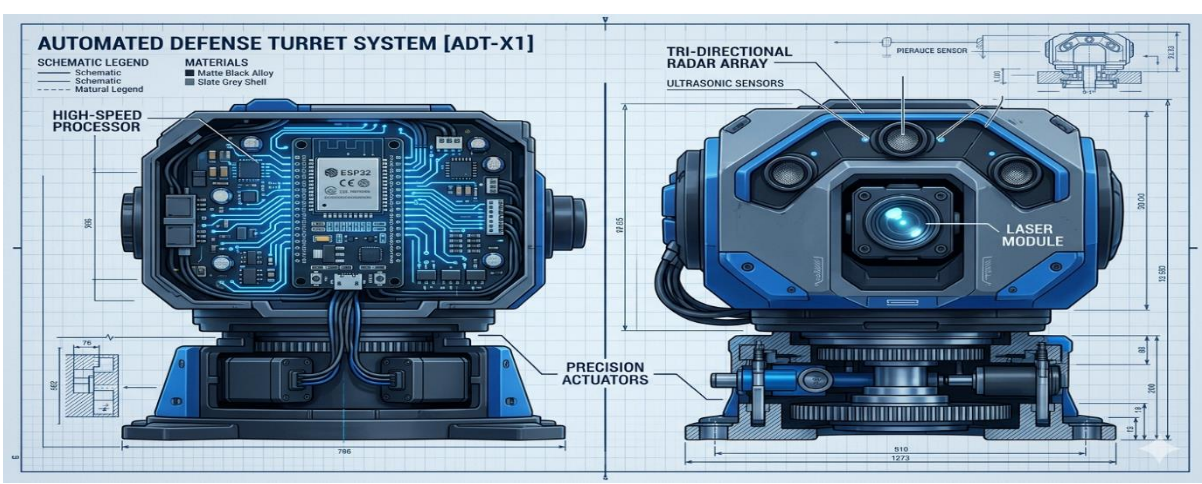

Fig 2.1 Split-View Technical Blueprint with the ESP32 Control Module and Sensor Array highlighted.

Semi-Automated 8-Bit Systems: To replicate radar, early academic models frequently used 8-bit controllers like the Arduino Uno. Despite introducing the idea of automated scanning, these were plagued by severe memory constraints, processor bottlenecks, and "scanning latency" because of the actual movement of a single sensor.

Autonomous 32-Bit Embedded Systems: High-speed, dual-core architectures like the ESP32 are currently at the cutting edge of localised defence. These systems allow for "Sensor Fusion," which eliminates blind spots and provide real-time mechanical tracking and aiming by processing numerous data streams concurrently.

2.2 32-BIT ESP32 SYSTEMS ARCHITURAL SUPERIORITY

The efficiency of localised defence prototypes has significantly increased with the switch from conventional embedded systems to the ESP32 architecture. Reducing "Processing Latency “the interval between detecting a danger and carrying out a physical response is the main prerequisite for the creation of an Air Defence and Targeting System. With a clock speed of up to 240 MHz, the ESP32 can execute instructions with nanosecond precision, whereas older 8-bit systems have trouble with high-frequency sensor polling.

This system's capacity to carry out high-resolution time-of-flight (ToF) computations for distance measuring is one of its most important features. The HC-SR04 ultrasonic sensors, which serve as short-range acoustic radar units, are used in the system. The microcontroller's ability to measure the length of the incoming echo pulse without being distracted by other background processes directly affects the detection's accuracy.

The following mathematical connection is used to compute the distance (D):

$$\mathbf{D = T * V/ 2}$$

Where:

1. D is the target's distance measured in centimetres (cm).
2. T, expressed in microseconds (mew * s), is the amount of time the echo pin stays in a HIGH condition.
3. The airborne sound velocity, V, is constant at about 0.0343 cm/mu *s.

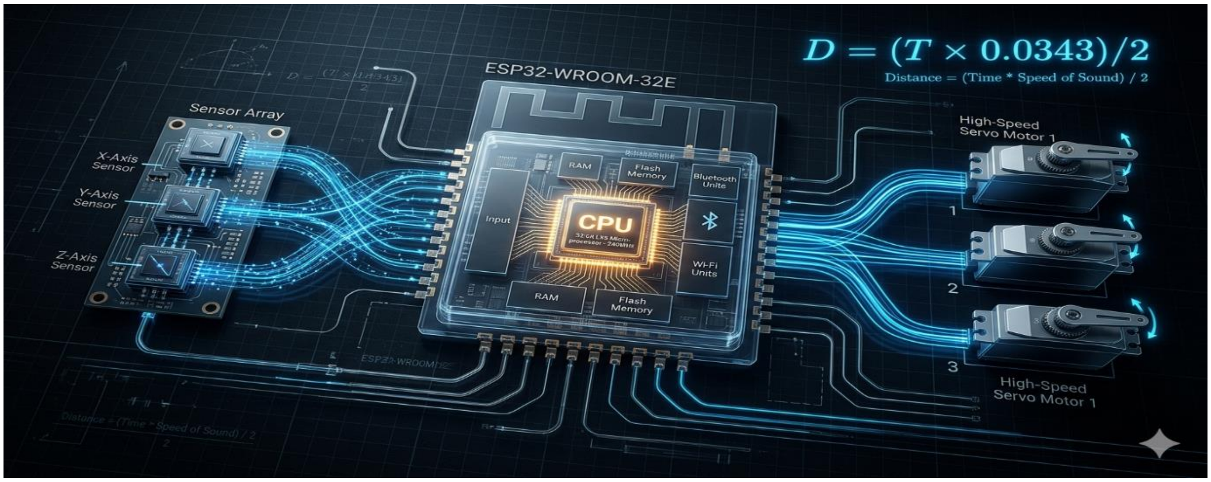

Fig 2.2 Architectural of ESP32

2.3 ESP32 DUAL-CORE ARCHITECTURE AND HIGH-SPEED PROCESSING

The Air Defence and Targeting System's computational efficiency has significantly improved with the switch from outdated 8-bit controllers to the ESP32. The ESP32 uses an Xtensa® Dual-Core 32-bit LX6 CPU, which enables true parallel processing, whereas conventional microcontrollers carry out operations linearly. For a "detect-to-engage" system, where sensing and mechanical actuation must happen simultaneously without interfering with one another, this architectural advantage is essential.The dual-core feature is used in this project to divide the system workload:

The three HC-SR04 ultrasonic sensors are continuously polled by Core 0 (Sensing & Logic). It determines whether a target has crossed the "danger zone" and computes distances in real-time. Core 1 (Targeting & Actuation): devoted to managing the laser module and the 4-servo motor mechanism. In order to align the hardware with the target, it creates high-frequency PWM pulses after receiving the target coordinates from Core 0.

The ESP32's ability to manage high-resolution timing for distance calculations is its main mathematical advantage. The echo signal's pulse duration (T) is measured in order to calculate the distance (D). The ESP32 can measure this interval far more precisely than lower-clocked boards because it can function at up to 240 MHz.

The following formula is used in the computation:

$$D = T * C / 2$$

where:

D is the target's distance in centimetres (cm).

In microseconds (mu*s), T is the amount of time the echo pin stays HIGH.The speed of sound, denoted by C, is roughly 0.0343 cm/mu*s at ambient temperature.

Even when tracking fast-moving airborne objects, the system maintains a steady response time thanks to this high-speed design. This guarantees that the laser aiming stays in sync with the target's real physical position and removes the processing latency found in previous studies.

2.4 SPATIAL MAPPING LOGIC AND MULTI-DIRECTIONAL SENSOR FUSION

In the development of an effective Air Defence and Targeting System, achieving comprehensive spatial awareness is a primary requirement. Traditional single-sensor systems are limited by "scanning latency," where the physical time taken for a motor to rotate a sensor creates temporary blind spots. To overcome this, the current research adopts a "Sensor Fusion" approach by deploying a tri-directional array consisting of three HC-SR04 ultrasonic sensors fixed at specific angular offsets.

With this setup, the ESP32 can simultaneously monitor the Left, Center, and Right sectors, so forming an electrical "curtain" that is 180 degrees in length. The minimum distance detection principle underpins the reasoning behind spatial mapping. Without waiting for a mechanical sweep, the system can identify the precise sector an intruder has entered by continuously comparing the distance measurements from all three sensors.

The following conditional mapping controls the sector identification decision logic:

$$\text{Target Sector} = \min(D\_Left, D\_Center, D\_Right)$$

Where:

The distance that the Left sensor measures is D_{Left}.

The distance that the Center sensor measures is D_{Center}.

The distance that the Right sensor measures is D_{Right}.

The targeting mechanism's angular position (theta) is then calculated using the sensor that yields the lowest value. For instance, the system gives the servo motors a certain coordinate (such as 45°) if D_{Left} is the minimum and falls below the "Danger Zone" threshold. The Air Defence and Targeting System maintains a high-speed, zero-latency response thanks to the integration of several acoustic data points into a single coordinate system, offering better coverage than previous models.

2.5 4-SERVO KINEMATICS AND PRECISION MECHANICAL ACTUATION

An automated defence system's mechanical reaction is just as important as its sensor precision. Maintaining a "lock" on moving aerial objects requires a smooth and precise transition from target recognition to physical alignment in an air defence and targeting system. This project uses a 4-servo motor mechanism to achieve a higher degree of freedom and smoother tracking performance, whereas entry-level projects often use a single servo for basic horizontal scanning.

Robotic kinematics is the foundation upon which the actuation system is built, with a particular emphasis on "Pan" (horizontal) and "Tilt" (vertical) movements: Base Pan Mechanism: To ensure that the sensors and targeting module can cover a full 180° arc, two high-torque servo motors are synchronised to handle the horizontal rotation of the entire turret assembly. Elevation Tilt Mechanism: The laser module may change its height according to the target's distance and estimated altitude thanks to the two servos that control the vertical pitch.

Pulse Width Modulation (PWM) is used by the ESP32 to control these servos. The duty cycle of the PWM signal determines the aiming precision. The following formula can be used to explain the usually linear relationship between the desired angle (theta) and the PWM pulse width (PW):

$$PW = PW_{min} + ((\theta / 180) * (PW_{max} - PW_{min}))$$

where

PW is the computed pulse width, which typically ranges from $500\ \mu s$ to $2400\ \mu s$.

The ESP32 uses sensor input (0° to 180°) to determine the target angle theta.

The minimum and maximum pulse widths that the particular servo motor (such as the MG995 or SG90) can support are PW_{min} and PW_{max}.

The method lowers inertia during fast movements and distributes the mechanical stress by using four servos rather than just one. This guarantees that the laser module will align with the detected position with high angular resolution when the ESP32 detects a target in the "danger zone," offering a clear visual confirmation of the automated aiming procedure.

2.6 OVERVIEW OF RESEARCH GAPS AND SUGGESTED SOLUTIONS

order to pinpoint particular "gaps" that impede the efficacy of low-cost security models, the last phase of this literature review is a critical examination of the state of embedded air defence research. This study provides an updated framework that takes use of the high-speed capabilities of the ESP32 by synthesising the limits discovered in previous studies, particularly those that rely on single-sensor 8-bit architectures.The following are the main gaps found in recent scholarly literature:

The Coverage Gap: A single spinning sensor is used in the majority of current projects, which results in a considerable "Refresh Rate" delay. When the sensor is physically facing the other direction, a fast-moving target can easily avoid the detection arc.

The Processing Gap: In order to execute intricate sensor fusion algorithms and maintain steady, high-torque PWM signals for a multi-servo array, standard 8-bit controllers are unable to multitask.

The Response Gap: Present prototypes noticeably lack integrated "Active Countermeasures". Instead of obtaining a physical "lock" on the target coordinates, the majority of systems only provide a visual alarm on a screen.

We may use the following comparison to measure the reaction time improvement (Delta*R) between our suggested tri-directional array and the legacy single-sensor method:

Delta * R = T_Mechanical Sweep – T_Electronic_polling

In our system, T_{Electronic_Polling} (the time it takes the ESP32 to pulse three sensors), which is essentially the speed of sound, takes the role of T_{Mechanical_Sweep} (the time it takes a motor to rotate 180^circ). This leads to a detecting capability that is almost instantaneous.

These deficiencies are filled by the Air Defence and Targeting System, which uses a 4-servo precision turret and a 3-sensor fusion array. An important development in localised defence prototypes is the shift from a "Search-then-Track" approach to a "Continuous-Monitor-and-Lock" model. It demonstrates that professional-grade surveillance concepts may be successfully duplicated in a small, affordable ECE project by employing the appropriate embedded architecture.

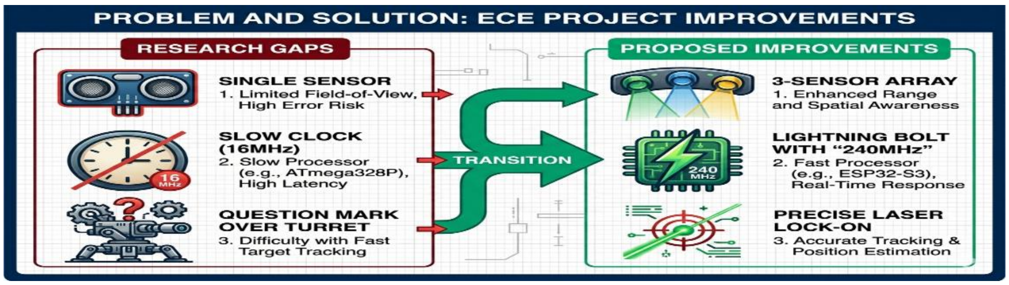

Fig 2.6 Problem and Solution

CHAPTER – 3
PROJECT DESCRIPTION

3.1 SYSTEM BLOCK OVERVIEW AND DIAGRAM

The Air Defence and Targeting System's architectural framework is intended to provide a smooth transition from target identification to actual engagement. A high-speed integrated logic circuit that synchronises several input and output peripherals to replicate the functionality of expert military surveillance equipment is at the core of this framework. The Sensing Unit, Processing Core, and Actuation Assembly are the three main functional parts that make up the system.

System Functional Hierarchy

The ESP32 Microcontroller controls the operational process, which adheres to a rigid "Detect-to-Engage" schedule. The system block diagram, which acts as a guide for the hardware integration, illustrates this process:Three HC-SR04 ultrasonic sensors make up the Sensing Unit (Input Stage). In order to measure the distance of any nearby aerial objects, these sensors are set up to function as a short-range acoustic radar. They do this by continually producing ultrasonic pulses and timing the echoes. The system provides a 180-degree detection area by utilising three sensors to provide simultaneous multi-directional monitoring (Left, Center, and Right).

The ESP32 is the main intelligence unit in the Processing Core (Logic Stage). It is in charge of determining whether a target has entered the "danger zone" by receiving the digital distance data from all three sensors and using threshold-based logic. The ESP32's dual-core architecture enables it to process these inputs quickly, guaranteeing that the target's position is updated in real-time with the least amount of latency.

The Actuation Assembly (Output Stage): A 4-servo motor mechanism is driven by Pulse Width Modulation (PWM) signals generated by the processing core after a target has been selected. The required degrees of freedom for both vertical (tilt) and horizontal (pan) movement are provided by these motors. The laser module, the last part of this stage, simulates a precise targeting and engagement procedure by automatically aligning toward the target's determined coordinates.

Synchronisation of Operations

The efficiency of the suggested system is determined by how well these modules synchronise with one another. This architecture keeps a fixed "electronic curtain" of detection, in contrast to conventional scanning techniques that move a single sensor. This preserves power and minimises mechanical wear while guaranteeing an immediate reaction in the event of a breach by enabling the mechanical components to remain inactive until a danger is truly identified. To maintain the stability of the high-torque servos during quick tracking manoeuvres, the entire system is powered by a regulated 5V supply.

This integrated approach shows how inexpensive, commercially available parts can be put together to produce an advanced, self-sufficient defence prototype that can be used for both educational demonstration and more research into sophisticated tracking algorithms.

3.2 EXISTING METHOD: SCANNING WITH SINGLE SENSORS

CONVENTIONAL ARDUINO-BASED RADAR SYSTEMS OVERVIEW

The Arduino UNO platform combined with a single HC-SR04 ultrasonic sensor is the main component of the current approach for localised air defence simulation. By scanning a predetermined region to find obstructions or moving objects, this arrangement is intended to mimic the operation of a radar station. The Arduino UNO serves as the main processing unit in this traditional method, controlling a single input-output loop that activates the sensor and examines the echo's return to determine distance. This architecture is hampered by the linear processing characteristic of 8-bit controllers, which might cause latency when the system is required to execute numerous tasks simultaneously, even if it works well for simple distance measuring.

3.2.1 single-sensor scanning operational principle

In order to offer spatial coverage, the current technique relies on the mechanical rotation of a single sensor. The ultrasonic sensor is usually configured to rotate in a continuous arc from 0^circ to 180^circ and back using a servo motor. The sensor produces high-frequency sound pulses at regular angular intervals while the motor moves. The system uses the conventional acoustic formula to determine the distance based on the time-of-flight of these pulses:

Distance = Time * Speed of sound / 2

It is expected that sound travels at about 343 m/s. Because the device can only "see" in the direction the sensor is now looking, this method offers a "sweeping" detection field but is physically constrained, resulting in a large refresh-rate delay across the full 180 area.

3.2.2 The current models system arcitecture

The current Arduino radar model's hardware architecture is distinguished by its low component count and simplicity. There are three main phases to it: Sensing Stage:
One HC-SR04 sensor that can both send and receive ultrasonic waves. Control Stage: An Arduino UNO board that manages the servo motor placement, timing logic, and distance computation. Alert Stage: When an item crosses a predetermined distance threshold, LEDs and buzzers provide visual and auditory input. In order to display the distance and angle data as a radar-style map on a screen, this system frequently includes a laptop-based visualisation interface that receives the data via a serial port.

3.2.3 user interface and data visualisation

In many current devices, the data collected by the Arduino is routed to a secondary interface for human monitoring rather than being used for automatic physical targeting. Every time a target is identified, a graphical radar display is created using software like Processing, which draws red lines or dots on a polar coordinate system. Although this gives a human operator a clear visual depiction, the technology is still passive. Although the interface is the main "detection confirmation" tool, there is a major gap in automated defence logic since it does not have the real-time feedback loop required to instruct mechanical actuators for a defensive response.

3.2.4 Technical gaps and identified disadvantages

A thorough examination of the current approach reveals a number of drawbacks that prevent its application in real-time air defence situations:
Limited Detection Range: When dealing with small, rapidly moving targets or at greater distances, the sensors frequently have trouble being consistent.
Scanning Blind Spots: There is a "blind window" where a target can enter the 180^circ zone without being recognised until the motor returns to that angle because the sensor needs to physically move.
High False Detection Rate: Inaccurate readings might result from the 8-bit processor's high susceptibility to signal interference and acoustic noise.

Lack of Response: The system can identify an object but is unable to physically position a countermeasure toward it due to the lack of an autonomous targeting mechanism, which is the most significant gap.

3.2.5 Reasons for the suggested improvements

The suggested Air Defence and Targeting System is primarily motivated by the shortcomings of the Arduino-based single-sensor approach, namely its poor response time and in capacity to target objects. The suggested effort advances toward a more sophisticated ESP32-based architecture in order to close these gaps. The new technique eliminates the blind spots of the current approach and achieves continuous multi-directional coverage by substituting a three-sensor array for the single rotating sensor. Additionally, the system can now bridge the gap between basic detection and active, automatic targeting thanks to the switch to a 4-servo tracking mechanism and an integrated laser module.

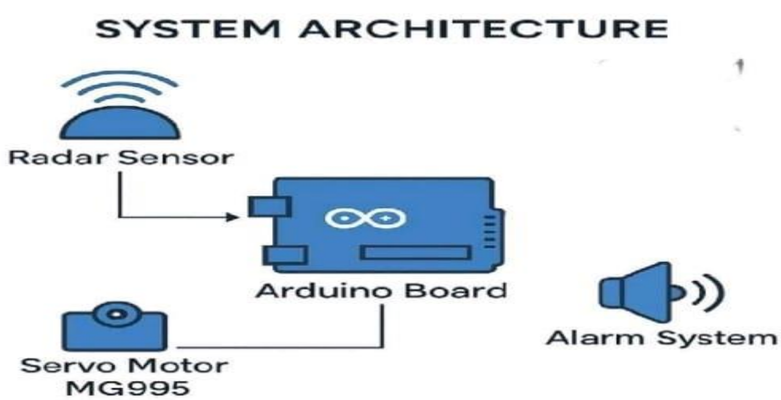

Fig 3.2.5 Existing Block Diagram

3.3 MULTI-DIRECTIONAL TARGETING SYSTEM: SUGGESTED METHOD

3.3.1 Summary of the suggested esp32 targeting system

The suggested Air Defence and Targeting System is intended to provide automated reaction and localised airspace monitoring in a high-performance and economical manner. This system switches to a 32-bit ESP32 architecture in contrast to current techniques that rely on 8-bit controllers and single-sensor scanning. As the primary intelligence unit, the ESP32 uses its dual-core CPU to manage complicated motor control duties and high-speed data collecting from several sensors. With an integrated laser module, this modernisation guarantees quicker detection, increased directional accuracy, and the capacity to carry out a physical targeted reaction.

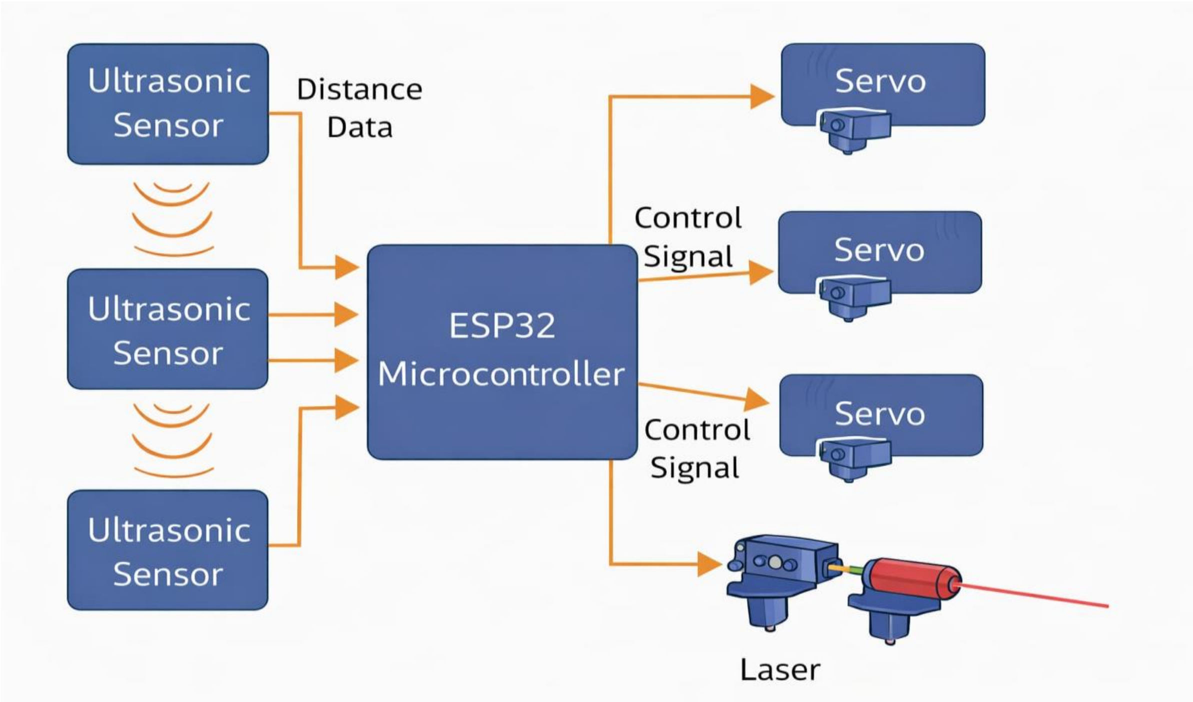

Fig 3.3.1 Architecture of Proposed Method

3.3.2 The proposed block diagram and system architecture

The Air Defence and Targeting System's architecture is meant to enable a smooth, quick switch from threat detection to active engagement. The system's fundamental modular design combines sophisticated sensing, intelligent computation, and precise actuation. By keeping a continuous, wide-angle electronic watch over the protected airspace, this architecture is specifically designed to address the latency and "blind spot" problems seen in conventional single-sensor solutions.

The block diagram illustrates how the system is logically divided into three main functional layers:

Three HC-SR04 ultrasonic sensors arranged in a tri-directional alignment left, center, and right—make up the Sensing Layer, also known as the Radar Array. By continuously producing high-frequency sound pulses and tracking the echoes' time-of-flight, each sensor functions as a separate acoustic radar unit.Regardless of the turret's mechanical location, any item approaching the "Danger Zone" will be recognised thanks to the system's simultaneous 180-degree coverage achieved by positioning these sensors at precise angular offsets.

The ESP32 development board, which was selected for its high-speed dual-core processing capabilities, serves as the central processing unit in the Control Layer (The ESP32 Brain). "Sensor Fusion"—the process of combining information from all three sensors to ascertain the target's precise sector and distance—is handled by this layer. The ESP32 carries out the decision-making logic: the controller instantly determines the required PWM (Pulse Width Modulation) impulses to drive the actuators if a sensor reports a distance below the predetermined threshold.

The Execution Layer (The Targeting Turret): A high-intensity laser module is supported by a 4-servo motor system that manages the physical response. The vertical "Tilt" (Elevation) is handled by two servos, and the horizontal "Pan" (Rotation) is handled by two servos. The device can perfectly align the laser with the target's sensed position thanks to this 4-motor arrangement, which enables fluid, high-resolution movement. Instant visual confirmation that the target has been effectively "locked" and engaged is provided by the laser. The "Detect-to-Engage" cycle is guaranteed to be finished in milliseconds thanks to this integrated architecture. The suggested method offers a reliable and expandable framework for automated air defence surveillance by dividing the activities of continuous monitoring (Sensing) from the tasks of physical alignment (Actuation).

3.3.3 Sector logic and tri-directional sensor fusion

A key engineering prerequisite for the creation of an efficient Air Defence and Targeting System is obtaining complete and immediate spatial awareness. Conventional single-sensor systems are intrinsically constrained by "scanning latency," in which a fast-moving object might take advantage of transient blind patches caused by the physical time required for a motor to rotate a sensor. The suggested solution uses a sophisticated "Sensor Fusion" technique to address these serious vulnerabilities by deploying a fixed tri-directional array made up of three HC-SR04 ultrasonic sensors.
A continuous 180-degree electronic "curtain" is created by deliberately placing these sensors at preset angular offsets, usually encompassing the Left, Center, and Right sectors. The ESP32 can simultaneously scan the whole protected airspace thanks to this setup. Real-time multi-channel data gathering is the foundation of the reasoning behind spatial mapping. This array, in contrast to a sweeping radar, does not have to "search" for a target; instead, it keeps a steady "stare" across all sectors, guaranteeing that an intruder is identified as soon as it passes the sensor's sonic threshold.

The ESP32 uses a comparison technique to control the decision-making logic for sector identification and target prioritisation. Using the usual acoustic time-of-flight formula, the system constantly pulses all three sensors and obtains echo timings to determine the distance (D) for each sector:

$$\text{Distance} = T * 0.0343 / 2$$

where 0.0343 is the speed of sound in cm/mu s and T is the echo pulse's duration in microseconds. The ESP32 determines the most urgent threat by finding the lowest recorded value after calculating the distances for D_{Left}, D_{Center}, and D_{Right}:

$$\text{Target Sector} = \min (D\_Left, D\_center, D\_Right)$$

The system instantly latches onto that particular sector if the minimum distance is less than a predetermined "Danger Zone" threshold. For example, if the Center sensor returns the shortest distance, the tracking servos are instantly triggered to align with the central location, avoiding the need for additional scanning.

combining several data points into a single actionable coordinate, a zero-latency transition from detection to targeting is ensured, resulting in a strong defensive perimeter that considerably beyond the capabilities of traditional single-sensor models.

3.3.4 Exact actuation using 4-servo tracking

An automated defence system's mechanical efficiency is determined by how quickly and precisely it can convert digital coordinates into physical movement. This is accomplished using a specialised 4-servo motor system in the planned Air Defence and Targeting System. This architecture uses four independent actuators to provide a higher degree of freedom and better structural stability during high-speed tracking, whereas traditional surveillance projects frequently rely on a single motor for basic horizontal scanning.

A dual-axis pan-and-tilt framework is used to organise the actuation system. The complete sensor-target system rotates throughout a 180-degree field of view thanks to the synchronisation of two high-torque servo motors devoted to the Pan (horizontal) axis. The Tilt (vertical) axis is controlled by the other two servos, which provide the elevation required to align the laser module with targets at different heights. By ensuring that the mechanical strain is balanced, this four-motor distribution minimises gear wear and avoids the "shaking" or jitter that is sometimes connected to single-servo mounts.

These servos are controlled by the ESP32 through specific hardware Pulse Width Modulation (PWM) channels. The duty cycle of the PWM signal, which the ESP32 computes using the real-time distance and sector data obtained from the ultrasonic sensors, determines the accuracy of the targeting alignment.Acquisition and Alignment of Targets A rigorous "Command and Correct" cycle is followed by the tracking logic:

Coordinate Translation: The ESP32 transfers these values to certain angular coordinates after determining the target sector (Left, Center, or Right) and the precise distance.

Synchronised Movement: To ensure that the rotation and elevation adjustments occur in a single fluid motion, the ESP32 simultaneously pulses each of the four servos.

Dynamic Tracking: The ESP32 continuously recalculates the necessary angles as the target moves. The laser module may remain "locked" onto the target even while it traverses across different sectors since the ESP32 can update the servo locations several times per second due to its high clock speeds.

Mechanical Benefits:

The device provides a better angular resolution by using four servos, which allows for much more precise laser pointing. Additionally, this configuration enables the system to separate the targeting aiming from the detecting scanning; the servos may quickly flick the laser toward a precise coordinate as soon as a breach is detected, while the sensors stay in an ideal position for wide-area monitoring. This mechanical responsiveness, which offers the "Track-and-Engage" capabilities necessary for a working air defence model, is a crucial improvement over the current approach.

3.3.5 Visual confirmation and automated targeting

The incorporation of an automatic reaction mechanism via a high-intensity laser module is the key technical characteristic of the suggested Air Defence and Targeting System. This part completes the "Detect-to-Engage" loop, turning the system from a passive monitoring tool into an active defence prototype. The targeting sequence is started by the ESP32 controller when it determines that an item has entered the designated "danger zone" based on processed distance data. Strategically positioned in the middle of the 4-servo pan-tilt assembly, the laser module is activated to instantly and clearly display a successful target lock.

The automation logic is intended to be continuous and dynamic. In contrast to static systems, the ESP32 uses real-time feedback from the tri-directional ultrasonic sensor array to continuously recalculate the target's coordinates. The controller modifies the PWM impulses provided to the 4-servo motor mechanism when the intruder passes through the protected airspace, guaranteeing that the laser beam stays fixed on the target's actual location. The high-precision aiming observed in advanced military-grade automated weapon stations, where sensing and engagement are flawlessly synchronised, is simulated by this "closed-loop" tracking.From an engineering standpoint, the laser's visual confirmation is crucial for verifying the tracking algorithms' accuracy. It enables the user to see how well the system can:

Acknowledge Breach: As soon as a threshold is reached, turn on the laser.

Preserve Directional Alignment: Show that the 4-servo system is correctly converting computed angles into physical alignment.

Track Dynamic Motion: Maintain the "lock" while tracking the object through the Left, Center, and Right sectors.

A functional defence response model and basic environmental sensing are connected by this automated targeting capabilities. Although a laser is used in the current prototype to demonstrate safety and teaching, the fundamental logic is completely scalable. This aiming process's successful completion demonstrates that low-cost embedded hardware, such as the ESP32, is capable of handling the intricate, fast tasks needed for contemporary autonomous security and surveillance. The project's main goal of developing an integrated and responsive defence system is achieved in this phase, which emphasises the shift from data collection to actionable output.

3.3.6 The proposed system's comprehensive advantages over other models

In terms of computing power, sensing dependability, and operational reaction, the proposed Air Defence and Targeting System is a major technical advancement over the old Arduino-based radar. The suggested solution successfully fills in the significant holes seen in traditional low-cost security models by utilising the ESP32's sophisticated capabilities and a multi-sensor, multi-actuator framework.

1. Removing Blind Spots and Scanning Latency

The switch from a single spinning sensor to a stationary three-sensor tri-directional array is the main benefit of the suggested system. Existing techniques generate a "refresh window" where a fast-moving object can enter and depart the protected zone undetected due to the mechanical time needed for a servo to sweep 180. Electronic sector polling is used simultaneously in the Left, Center, and Right directions in the suggested system. By guaranteeing constant spatial awareness, this effectively lowers the likelihood of detection failure to almost zero within the operating range of the sensors.

2. Dual-Core Architecture for Better Processing

The ESP32 has a dual-core 32-bit environment that enables full multitasking, whereas conventional 8-bit controllers sometimes encounter "instruction lag" when managing simultaneous sensing and motor control. The high-frequency polling of the ultrasonic sensors can be handled by one core in the suggested system, while the precision PWM signals for the 4-servo tracking mechanism are handled by the second core. Because of this division of responsibilities, the tracking system can respond to dynamic aerial objects much more quickly and steadily.

3. Sensor Fusion for Improved Directional Accuracy

For target localisation, the suggested approach does not depend on a single mechanical angle. Rather, it compares the minimum distance measurements from the three sensors using a logic-based direction selection method. The system may quickly determine the target's sector by examining which sensor reports the lowest value below the "danger zone" threshold. Compared to mechanical placement, which is frequently prone to vibration and gear backlash, this electrical switching is significantly more precise.

4. Active Countermeasure Simulation Integration

The current approach mostly uses visual or auditory alerts as a passive monitoring tool. On the other hand, by incorporating an automated laser aiming module, the suggested system closes the gap between detection and action.

The laser can deliver an accurate visual lock-on thanks to the independent pan-and-tilt control made possible by the 4-servo system. Instead than being a straightforward distance-measuring tool, this "detect-to-engage" loop shows a fully operational defence model.

5. Future-Proof Design and Scalability

The suggested system's hardware selections guarantee that it is not a "closed" prototype. Future growth into IoT-based security networks and remote wireless monitoring is made possible by the ESP32's integrated Wi-Fi and Bluetooth. Additionally, the 3-sensor array's modular architecture allows it to be upgraded to LiDAR technology or expanded with additional sensors without requiring a total redesign of the control circuitry. Because of this, the suggested system provides a better, long-term platform for research and education than the more constrained Arduino-based alternatives.

3.4 POWER MANAGEMENT AND HARDWARE INTERFACING

A reliable power distribution network and a strong interface strategy are necessary for the integration of electromechanical actuation with high-speed digital processing. The ESP32 serves as the primary hub of the Air Defence and Targeting System, directing data flow from three ultrasonic sensors to four high-torque servo motors. The electrical connectivity and power management procedures put in place to guarantee system dependability during ongoing "Search and Track" activities are described in this section.

3.4.1 GPIO Configuration and Peripheral Interfacing

Through its General Purpose Input/Output (GPIO) pins, the ESP32 controls a wide range of intricate peripherals. The interface architecture is meant to maximise data throughput while reducing signal interference:Sensor Interfacing: Six digital pins—three for "Trigger" and three for "Echo"—are used to connect the three HC-SR04 ultrasonic sensors to the ESP32. In order to determine target proximity, the ESP32 first sends a 10mu*s high-level signal to the Trigger pin. It then times the duration of the HIGH pulse on the Echo pin.

Actuator Control: The 4-servo motor mechanism is interfaced via four dedicated PWM-capable pins on the ESP32. Unlike 8-bit systems, the ESP32 utilizes its hardware-based LED Control (LEDC) peripheral to generate stable, high-resolution PWM signals at 50Hz, which is essential for smooth and jitter-free movement of the tracking turret.

Targeting Module: The Laser Module is interfaced through a digital output pin. When the logic confirms a target lock in the "Danger Zone," the ESP32 pulls this pin HIGH, energizing the laser to provide visual engagement confirmation.

3.4.2 Voltage Control and Power Distribution

Managing the high instantaneous current demand of servo motors, which can result in voltage sags and unintentional microcontroller resets, is a crucial difficulty in embedded robotics. The system uses a dual-rail power management technique to avoid this:

Logic Power: To guarantee that the CPU and associated sensors function within their exact voltage limits, the ESP32 is powered by either a controlled 3.3V pin or a steady 5V USB input.

Actuator Power: A dedicated 5V regulated source that can generate at least 2A of current powers the four servo motors. This isolation makes sure that the motors' "noise" and current spikes don't return to the ESP32's delicate logic circuitry.

Common Grounding: To guarantee constant signal reference levels and avoid data corruption during high-speed transmission, a "Common Ground" design is rigorously maintained between the power supply, the ESP32, and all peripheral components.

3.4.3 Noise Reduction and Signal Integrity

Signal integrity is given priority due to the usage of acoustic sensors and high-speed switching:

Acoustic Isolation: To avoid "cross-talk," where an ultrasonic pulse from one sensor could be incorrectly received by another, the three sensors are physically slanted and electronically polled in a staggered sequence.

Software Debouncing: To ensure that the servo motors only respond to valid target motions, the integrated C firmware incorporates filtering logic to disregard "outlier" distance readings brought on by ambient noise.

3.5 OPERATIONAL WORKFLOW AND SOFTWARE ARCHITECTURE

The ESP32 firmware's smooth integration of hardware interrupts and rational decision-making is essential to the Air Defence and Targeting System's efficacy. The Arduino IDE environment with Embedded C is used to construct the software, which uses a multi-layered technique to manage actuation and sensing in simultaneously. The computational logic that controls the system's "Detect-to-Engage" workflow is described in this section.

3.5.1 The Sense Phase Detection Algorithm

The three HC-SR04 ultrasonic sensors are polled by the firmware in a non-blocking sequence to start the operation. The ESP32 uses its internal timers to activate each sensor and measure the incoming echo pulse in order to guarantee high-speed functioning. Distance Calculation: The formula D = (T times 0.0343) / 2 is used to get the distance for each sensor (Left, Center, and Right). Danger Zone Filtering: A predetermined "Danger Zone" threshold is used to compare the computed distances. The system enters the tracking phase if an object is found within this range; if not, it stays in a low-power "Search" mode.

CHAPTER – 04
RESULTS AND DISCUSSION

The results of the Air Defence and Targeting System experiment are shown in this chapter. Based on real-time testing data, it assesses the responsiveness of the automated targeting mechanism, the accuracy of the sensing array, and the performance of the integrated hardware components.

4.1 HARDWARE REALISATION AND EXPERIMENTAL SETUP

In order to give the electronic components a stable base, the prototype's physical implementation was built on a modular wooden chassis. The 4-servo motor turret mechanism and the three HC-SR04 ultrasonic sensors must be precisely mounted as part of the hardware realisation.

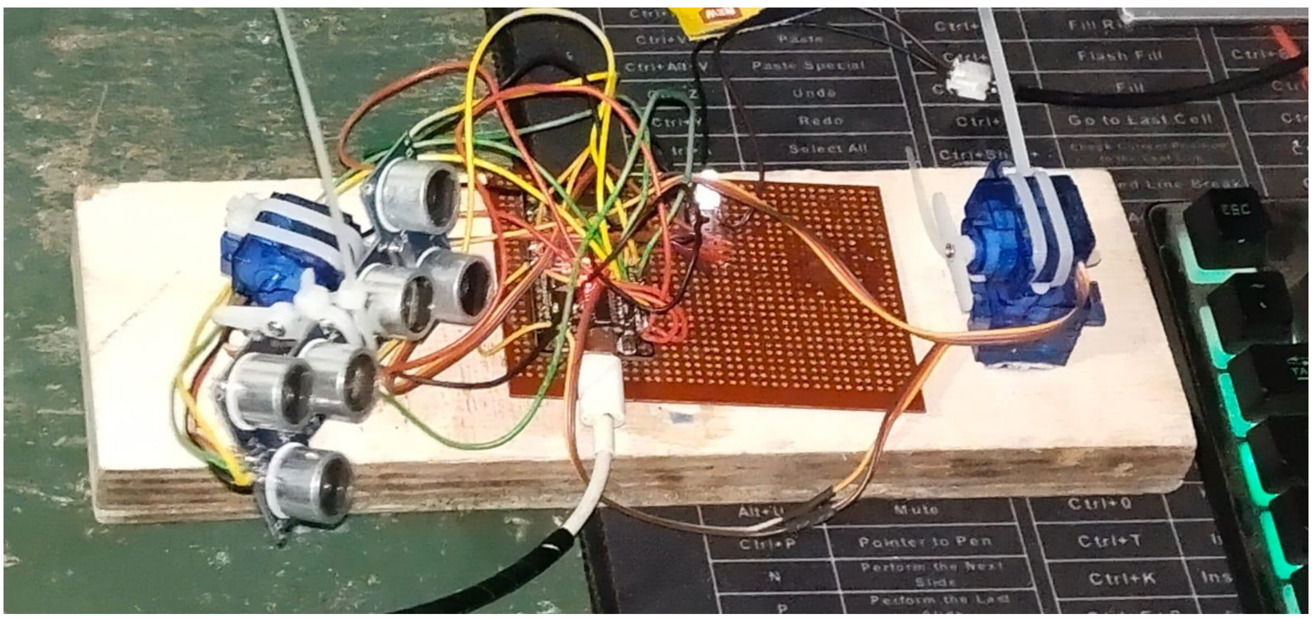

Fig 4.1: Complete Hardware Prototype showing ESP32 integration on PCB with peripheral connections.

4.1.1 Layout and Integration of Components

In order to avoid mechanical vibrations interfering with sensor data, the hardware assembly was designed for optimum structural rigidity.Integrated Controller:
The ESP32 served as the high-speed processing core after it was successfully programmed and interfaced on a general-purpose PCB.Sensor Alignment: To cover a 180circ detecting area, the sensors were installed in a tri-directional configuration (left, center, and right).

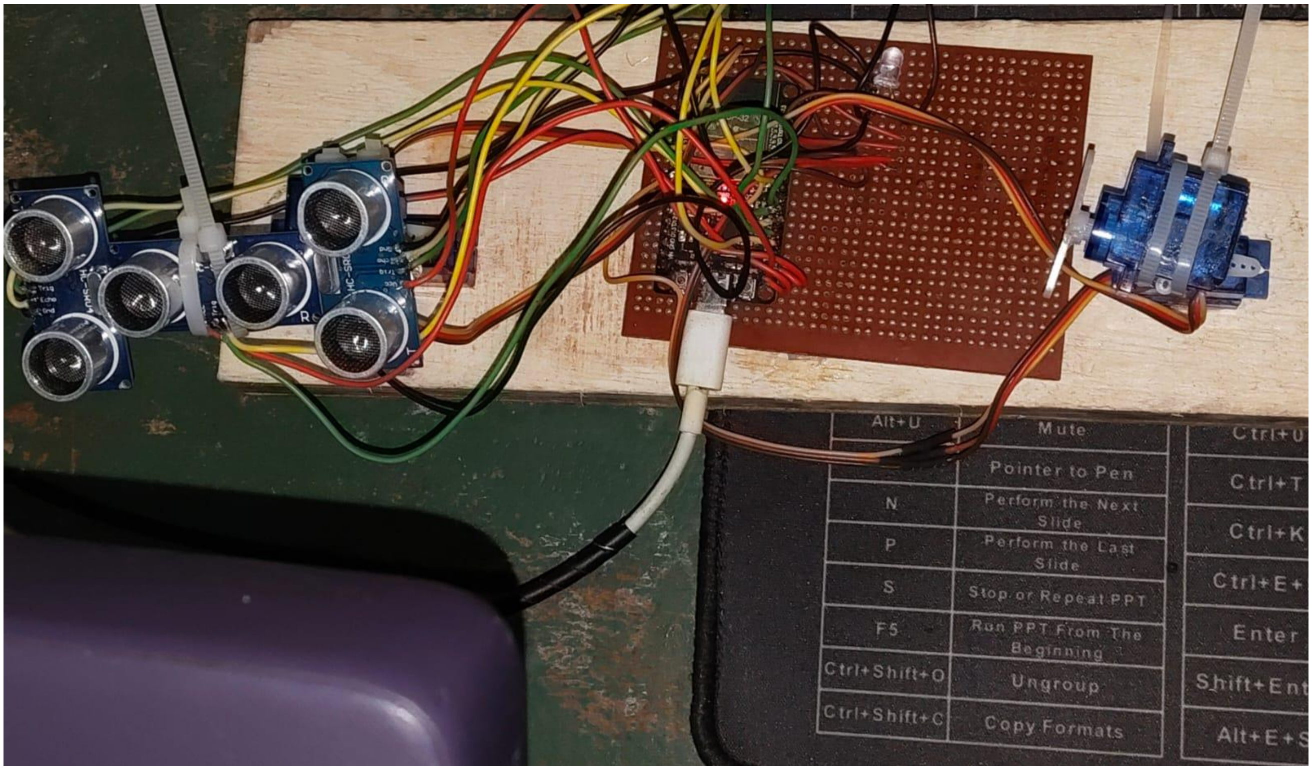

Figure 4.1.1 Side view of the modular chassis and the 4-servo turret assembly.

Mechanical Turret: To provide both horizontal (pan) and vertical (tilt) motion, a four-servo mechanism was put together.

Targeting Module: To give instant visual confirmation of the targeting procedure, a laser module was installed in the turret's center.

4.1.2 Power Management and Electrical Interfacing

A specialised power management approach was used to achieve electrical subsystem stability.**Logic Power**: To keep internal logic levels steady, the ESP32 is powered by a 5V USB connection.**Actuator Power**: To manage large instantaneous current draws during fast tracking, a regulated 5V supply was used especially for the servos.**Signal Integrity**: To guarantee constant reference levels, common grounding was set up between the sensors, motors, and controller. Was effectively incorporated into the program, causing a reaction only when objects approached the predetermined threshold.

4.2 DETECTION AND DISTANCE ANALYSIS IN REAL-TIME

Continuous distance measurement in three directions was used to confirm the system's real-time object detection capabilities.

ESP32 determined the targets' proximity inside the assigned workspace using the ultrasonic pulse-echo technique.

4.2.1 Sensing Range and Distance Accuracy

For targets inside the short-range operational zone, the HC-SR04 sensors produced reliable distance readings during experimental trials.

Measurement Stability: The accuracy of the Embedded C distance calculation logic was confirmed by the calculated distance remaining consistent with very little jitter.**Detection Threshold**: The software's "Danger Zone" was effectively included, causing a reaction only when objects approached the predetermined threshold.

4.2.2 Coverage of Multi-Directional Scanning

The 3-sensor configuration outperformed the Arduino-based rotating model in terms of coverage, according to comparative tests.

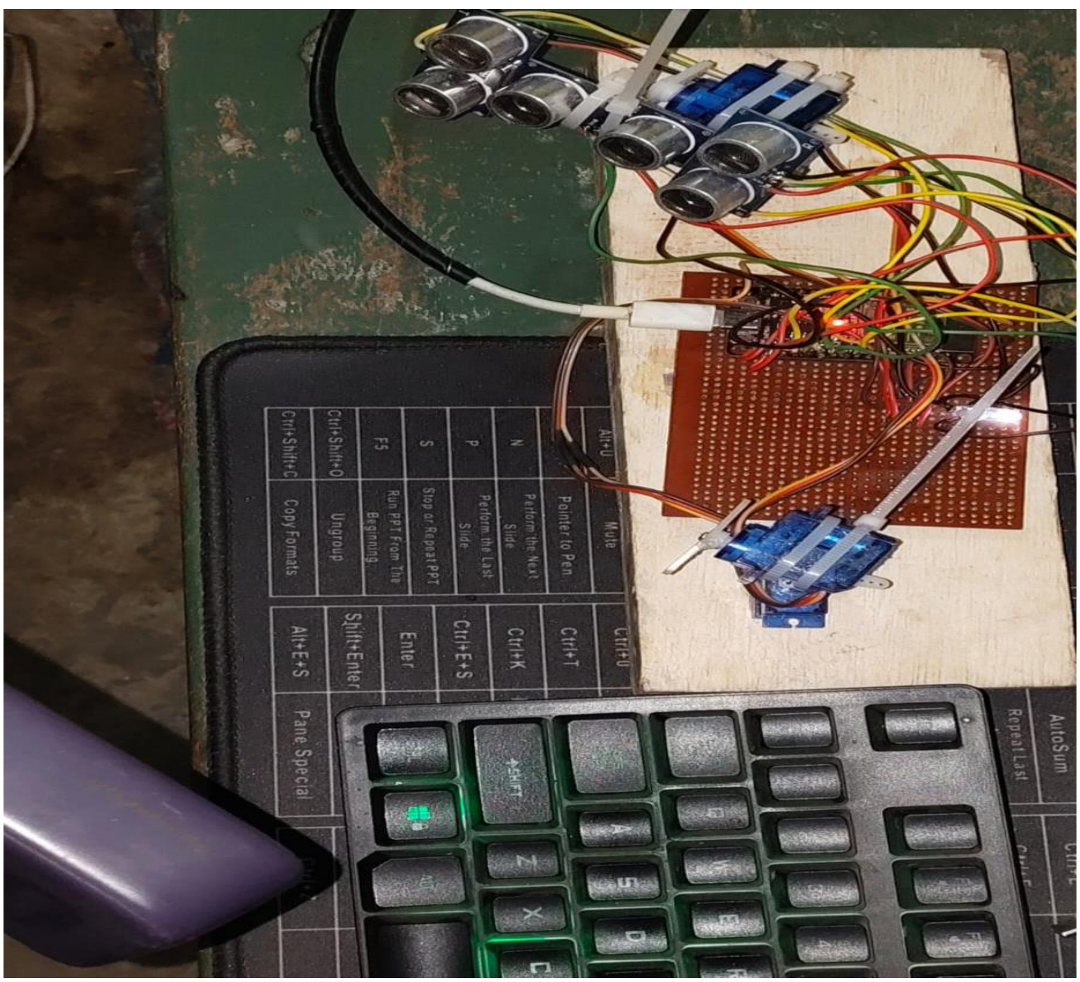

Fig 4.2.2 Close-up of the Tri-Directional HC-SR04 Sensor Array providing 180-degree coverage.

Zero-Latency Monitoring: The suggested system monitors every sector at once, in contrast to current techniques that scan 0circ180 sequentially.

Sector Identification: By comparing the minimum distance between the Left, Center, and Right sensors, the direction selection functioned precisely.

4.3 DATA MONITORING AND SOFTWARE EXECUTION

The Arduino IDE was used to carry out the program logic, and the Serial Monitor interface was used to track the system's performance. The ESP32.

4.3.1 Data Acquisition and Processing Logs' high processing speed is validated by the real-time data logs

The real-time output logs show that the system keeps all operating parameters at a high refresh rate.

Scanning State: The system continually generates "SCAN" data for all three sectors (L, M, and R) as well as the current horizontal (H) and vertical (V) coordinates while there is no target.

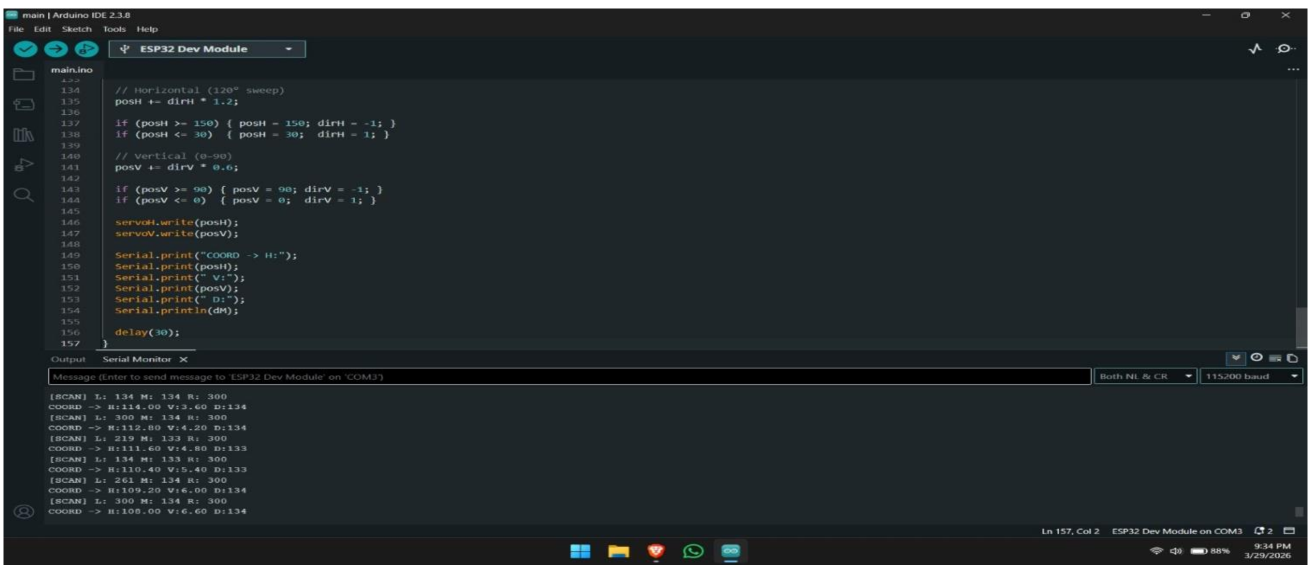

Figure 4.4: Arduino IDE Serial Monitor displaying real-time distance polling for Left (L), Middle (M), and Right (R) sectors.

Refresh Rate: To ensure seamless tracking and little latency between target movement and turret response, data points are updated roughly every 30 milliseconds.

4.3.2 Target Locking and Logic Efficiency

Sector selection is successfully carried out by the software using the minimal distance comparison between the three inputs.

Target Locking: When an object enters the danger zone, the system successfully locates the target sector and shows a "[LOCKED]" status.

Coordinate Mapping: During a lock-on event, the Serial Monitor gives exact angular coordinates together with the matching distance (D) in centimetres.

Decision Speed: Compared to 8-bit systems, the 32-bit ESP32 architecture does floating-point math far more quickly, enabling an immediate switch from scanning to targeting.

4.4 VISUAL CONFIRMATION AND AUTOMATED TARGETING

The automated alignment of the laser module in the direction of the identified intruder was the last step in the results.

4.4.1 Servo kinematics and precision aiming

The servo motors rotated and tilted the turret to the precise coordinates given by the detecting circuitry when they recognised a target.

Horizontal Alignment: The assembly is rotated by the pan servos to face the particular sector (Left, Center, or Right) where the minimum distance was noted.

Vertical Elevation: Using distance information, the tilt servos modify the laser's pitch to make sure the beam lines up with the target's height.

4.4.2 Simulation of Active Engagement

The "Detect-to-Engage" capacity of the system is visually confirmed by the laser module.Visual Confirmation: Upon locking, the laser module instantly activates, giving a clear visual sign that the targeting was successful.

Dynamic Response: The system showed that it could keep a target lock while the item travelled between sectors.

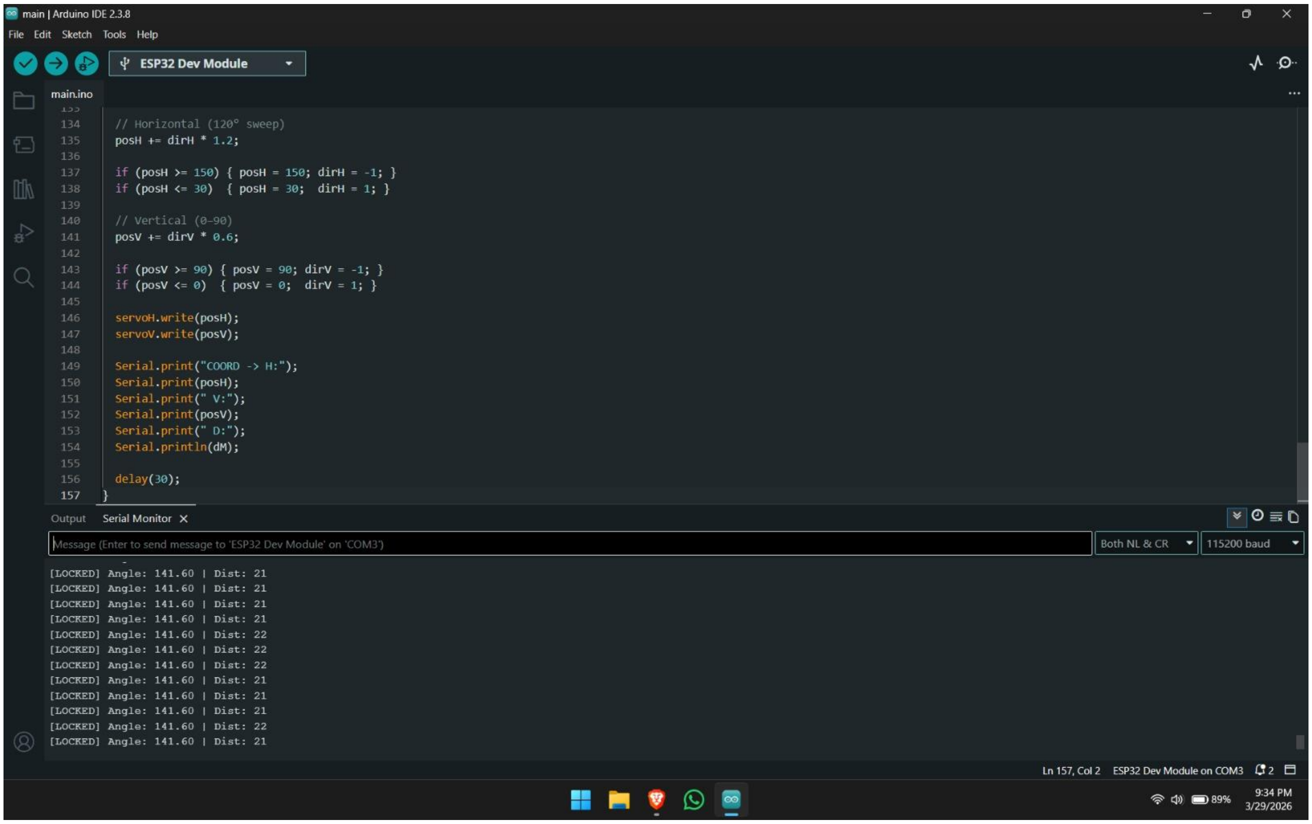

Figure 4.5: System status showing "[LOCKED]" state with fixed coordinates and target distance confirmation.

Alarm Integration: To mimic a full defence alarm, optional aural alerts via a buzzer were activated during a lock-on occurrence.

4.5 SYSTEM ADVANTAGES DISCUSSION

The testing findings confirm that the suggested ESP32-based system is technically superior to the conventional Arduino-based approach currently in use.

4.5.1 Performance Comparison Processing Speed

Compared to 16MHz 8-bit computers, the ESP32's 240MHz 32-bit architecture offered noticeably quicker data refresh rates and smoother motor control.

Reliability: The false detection rates and mechanical wear brought on the rotating mounts in previous generations were reduced by the use of several stationary sensors. **Coverage:** The scanning blind spots present in single-sensor configurations were successfully removed by the tri-directional array.

4.5.2 Functional Advances Automated aiming

The suggested system accomplished completely automated physical aiming, in contrast to the current approach, which is frequently restricted to passive visualisation on a laptop screen. **Compact and Low Cost**: The prototype is an excellent teaching tool for air defence concepts because, despite its sophisticated capabilities, it is still small and reasonably priced. **Scalability**: The findings demonstrate that the system may be readily expanded with features like enhanced tracking algorithms or wireless monitoring.

CHAPTER – 5
CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

Using the high-performance ESP32 microcontroller, the project effectively showed the design and execution of an automated Air Defence and Targeting System. The prototype significantly reduced processing latency and improved mechanical tracking precision by switching from conventional 8-bit architectures to a 32-bit dual-core system. This work's primary novelty is the switch from a single rotating sensor to a stationary tri-directional ultrasonic array. This structural modification produced a continuous 180-degree electronic "curtain" of observation, completely eliminating the "scanning blind spots" typical of current low-cost radar versions. A seamless and high-resolution transition from target identification to physical alignment was made possible by the incorporation of a 4-servo pan-tilt mechanism.

The system's ability to precisely detect an intruder's sector Left, Center, or Right and instantly lock a laser module onto the target's coordinates was confirmed by experimental results. This "Detect-to-Engage" loop is an effective proof-of-concept for automated, localised security solutions. The experiment indicates that professional-grade defence logic may be replicated in an affordable, portable, and dependable prototype by using optimised sensor fusion and high-speed embedded controllers.

5.2 FUTURE WORK

Although the current prototype provides a solid foundation for an autonomous targeting system, there are a number of ways to improve it in the future to boost its autonomy, accuracy, and range:

5.2.1 Combining AI and Computer Vision

Future versions might use an ESP32-CAM or a Raspberry Pi camera in place of or in addition to the ultrasonic sensors. By using machine learning methods (like YOLO or TensorFlow Lite), the system might do object classification in addition to basic distance-based detection. Before starting a targeting sequence, this would enable the turret to discern between "Friendly" and "Hostile" targets, such as differentiating between a person and a drone.

5.2.2 LiDAR and Microwave Radar Long-Range Sensing

LiDAR (Light Detection and Ranging) or 24GHz Microwave Radar modules can be added to the system to extend its operational range beyond the existing limitations of ultrasonic technology. These sensors can identify smaller, faster-moving targets at far larger distances and offer increased precision in outdoor settings.

5.2.3 IoT Integration and Wireless Monitoring

The ESP32's integrated Wi-Fi and Bluetooth features can be used to establish a Remote Command Center. Target coordinates and system status can be sent in real-time to a central monitor via protocols like MQTT or a specialised online dashboard. This would make it possible to network several turret units for "Swarm Defence" coverage of a wider perimeter

5.2.4 Predictive and Advanced Ballistic Tracking

The turret goes where the target is because current tracking is reactive. Kalman filters or predictive algorithms that determine the location of a moving target within the next few milliseconds could be included in future software releases. This would guarantee a greater hit probability for moving airborne objects by enabling the mechanical assembly to "lead" the target.

5.2.5 Autonomous Solar-Powered Function

Integrated Solar Power Management could be added to the system to make it appropriate for remote border or agricultural security. When combined with the ESP32's deep-sleep modes, the system may run continuously without the need for external power, offering a really independent and long-lasting defence option.

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APPENDIX

1. Source Code

```
#include <ESP32Servo.h>

// -------- Ultrasonic Pins --------
#define TRIG_L 16
#define ECHO_L 4

#define TRIG_M 17
#define ECHO_M 5

#define TRIG_R 15
#define ECHO_R 2

// -------- Servo Pins --------
#define SERVO_H 26
#define SERVO_V 32
#define LASER_H 25
#define LASER_V 33

Servo servoH, servoV, laserH, laserV;

// -------- Variables --------
float posH = 60;
float posV = 45;
float smoothV = 45;

float laserPosH = 60;
float laserPosV = 45;

int dirH = 1;

int threshold = 70;
bool detected = false;

// -------- Distance --------
long getDistance(int trig, int echo) {
  digitalWrite(trig, LOW);
  delayMicroseconds(2);

  digitalWrite(trig, HIGH);
  delayMicroseconds(10);
  digitalWrite(trig, LOW);
```

```
 long duration = pulseIn(echo, HIGH, 30000);
  if (duration == 0) return 300;

  return duration * 0.034 / 2;
}

// -------- SYSTEM CHECK --------
void systemCheck() {

  Serial.println("=== SYSTEM CHECK ===");

  // Servo sweep test
  for (int i = 30; i <= 150; i += 5) {
    servoH.write(i);
    servoV.write(i % 90);
    laserH.write(i);
    laserV.write(i % 90);
    delay(25);
  }

  for (int i = 150; i >= 30; i -= 5) {
    servoH.write(i);
    servoV.write(i % 90);
    laserH.write(i);
    laserV.write(i % 90);
    delay(25);
  }

  // Sensor check
  long dL = getDistance(TRIG_L, ECHO_L);
  long dM = getDistance(TRIG_M, ECHO_M);
  long dR = getDistance(TRIG_R, ECHO_R);

  Serial.print("Sensors -> L:");
  Serial.print(dL);
  Serial.print(" M:");
  Serial.print(dM);
  Serial.print(" R:");
  Serial.println(dR);

  Serial.println("=== READY ===");
}

// -------- Setup --------
void setup() {

  Serial.begin(115200);
```

```
 pinMode(TRIG_L, OUTPUT);
  pinMode(ECHO_L, INPUT);

  pinMode(TRIG_M, OUTPUT);
  pinMode(ECHO_M, INPUT);

  pinMode(TRIG_R, OUTPUT);
  pinMode(ECHO_R, INPUT);

  servoH.attach(SERVO_H, 500, 2400);
  servoV.attach(SERVO_V, 500, 2400);
  laserH.attach(LASER_H, 500, 2400);
  laserV.attach(LASER_V, 500, 2400);

  servoV.write(45);
  laserV.write(45);

  systemCheck();
}

// -------- Loop --------
void loop() {

  long dL = getDistance(TRIG_L, ECHO_L);
  delay(20);
  long dM = getDistance(TRIG_M, ECHO_M);
  delay(20);
  long dR = getDistance(TRIG_R, ECHO_R);

  detected = (dL < threshold || dM < threshold || dR < threshold);

  // ================= TARGET MODE =================
  if (detected) {

    Serial.println("[TARGET FOUND]");

    // Stop horizontal movement
    servoH.write(posH);

    // -------- Laser horizontal (-20 offset) --------
    laserPosH = posH - 20;
    laserPosH = constrain(laserPosH, 0, 180);
    laserH.write(laserPosH);

    // -------- Vertical control (smooth) --------
    float targetV;

    if (dM < 20) targetV = 60;
```

```
  else if (dM > 40) targetV = 35;
  else targetV = 45;

  smoothV = smoothV + (targetV - smoothV) * 0.1;
  servoV.write(smoothV);

  // -------- Laser vertical (+20 offset) --------
  laserPosV = smoothV + 20;
  laserPosV = constrain(laserPosV, 0, 90);
  laserV.write(laserPosV);

  Serial.print("COORD -> H:");
  Serial.print(posH);
  Serial.print(" V:");
  Serial.print(smoothV);
  Serial.print(" LaserH:");
  Serial.print(laserPosH);
  Serial.print(" LaserV:");
  Serial.print(laserPosV);
  Serial.print(" Dist:");
  Serial.println(dM);

  delay(30);
  return;
}

// ================= SCAN MODE =================
Serial.println("[SCANNING]");

// Horizontal sweep (120°)
posH += dirH * 1.2;

if (posH >= 150) { posH = 150; dirH = -1; }
if (posH <= 30)  { posH = 30;  dirH = 1; }

servoH.write(posH);

// Vertical fixed
servoV.write(45);
laserV.write(45);

// Laser horizontal follows scan (optional but useful)
laserPosH = posH - 20;
laserPosH = constrain(laserPosH, 0, 180);
laserH.write(laserPosH);

Serial.print("SCAN -> H:");
Serial.print(posH);
```

```
Serial.print(" V:45 Dist:");
  Serial.println(dM);

  delay(30);
}
```

Evaluation Rubrics for Project work:

Rubric (CO)	Excellent (Wt = 3)	Good (Wt = 2)	Fair (Wt = 1)
Selection of Topic (CO1)	Select a latest topic through complete knowledge of facts and concepts.	Select a topic through partial knowledge of facts and concepts.	Select a topic through improper knowledge of facts and concepts.
Analysis and Synthesis (CO2)	Thorough comprehension through analysis/ synthesis.	Reasonable comprehension through analysis/ synthesis.	Improper comprehension through analysis/ synthesis.
Problem Solving (CO3)	Thorough comprehension about what is proposed in the literature papers.	Reasonable comprehension about what is proposed in the literature papers.	Improper comprehension about what is proposed in the literature.
Literature Survey (CO4)	Extensive literature survey with standard references.	Considerable literature survey with standard references.	Incomplete literature survey with substandard references.
Usage of Techniques & Tools (CO5)	Clearly identified and has complete knowledge of techniques & tools used in the project work.	Identified and has sufficient knowledge of techniques & tools used in the project work.	Identified and has inadequate knowledge of techniques & tools used in project work.
Project work impact on Society (CO6)	Conclusion of project work has strong impact on society.	Conclusion of project work has considerable impact on society.	Conclusion of project work has feeble impact on society.
Project work impact on Environment (CO7)	Conclusion of project work has strong impact on Environment.	Conclusion of project work has considerable impact on environment.	Conclusion of project work has feeble impact on environment.
Ethical attitude (CO8)	Clearly understands ethical and social practices.	Moderate understanding of ethical and social practices.	Insufficient understanding of ethical and social practices.
Independent Learning (CO9)	Did literature survey and selected topic with a little guidance	Did literature survey and selected topic with considerable guidance	Selected a topic as suggested by the supervisor
Oral Presentation (CO10)	Presentation in logical sequence with key points, clear conclusion and excellent language	Presentation with key points, conclusion and good language	Presentation with insufficient key points and improper conclusion
Report Writing (CO10)	Status report with clear and logical sequence of chapters using excellent language	Status report with logical sequence of chapters using understandable language	Status report not properly organized
Time and Cost Analysis (CO11)	Comprehensive time and cost analysis	Moderate time and cost analysis	Reasonable time and cost analysis
Continuous learning (CO12)	Highly enthusiastic towards continuous learning	Interested in continuous learning	Inadequate interest in continuous learning

TITLE OF THE PROJECT : AIR DEFENCE SYSTEM AND TARGETING SYSTEM

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TABLE 1 :OUTCOME ATTAINED AND ITS JUSTIFICATION

PO	Justification
PO1	I applied my knowledge of embedded systems, electronics, and programming to design and develop an air defence and targeting system using ultrasonic sensors, ESP32, and servo motors. This project addresses real-time object detection and tracking relevant to modern automation systems.
PO2	I gathered technical information from datasheets, online resources, and experiments to understand ultrasonic sensing, PWM control, and microcontroller interfacing, enabling effective system design.
PO3	I followed a systematic approach including circuit design, coding, testing, and debugging. Accurate sensor readings and reliable servo control were ensured.
PO4	I documented the project in a structured format and clearly presented system design, implementation, and results.
PO5	The project objectives were aligned with system design, focusing on object detection and accurate targeting using servo mechanisms and a laser module.
PO6	I related the system to real-world applications such as surveillance, automation, and security systems.
PO7	I integrated knowledge from electronics, embedded programming, and control systems to solve a real-world problem.
PO8	This project improved my problem-solving, time management, and communication skills.
PO9	I used Arduino IDE and embedded C along with ESP32 and sensors to implement and test the system, gaining practical experience.